## 10.2: Matrix Representations of Graphs

**Definition.**
An $m \times n$ **matrix** $A$ **over a set** $S$ is a rectangular array of elements of $S$ arranged into $m$ rows and $n$ columns:

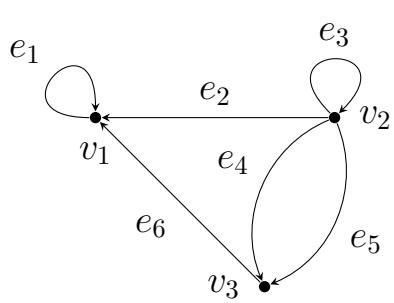
$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1j} & \dots & a_{jn} \\ a_{21} & a_{22} & \dots & a_{2j} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \dots & a_{ij} & \dots & a_{in} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mj} & \dots & a_{mn} \end{bmatrix}$$

*Note*: Two matrices $A$ and $B$ are only equal if every component of $A$ and $B$ are equal:

$$A = B \Leftrightarrow a_{ij} = b_{ij} \ \ \forall i = 1, 2, \dots, m, \text{ and } \forall j = 1, 2, \dots, n.$$

**Definition.**
Let $G$ be a directed graph with ordered vertices $v_1, v_2, \dots, v_n$. The **adjacency matrix of** $G$ is the $n \times n$ matrix $A = (a_{ij})$ over the set of nonnegative integers such that

$$a_{ij} = \text{ the number of arrows from } v_i \text{ to } v_j, \ \forall i, j = 1, 2, \dots, n.$$

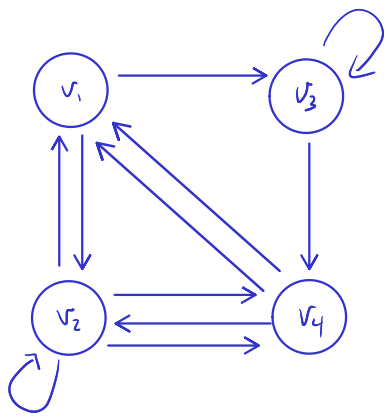



$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \end{array} \begin{array}{c} \begin{array}{ccc} v_1 & v_2 & v_3 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 2 \\ 1 & 0 & 0 \end{bmatrix} \end{array}$$

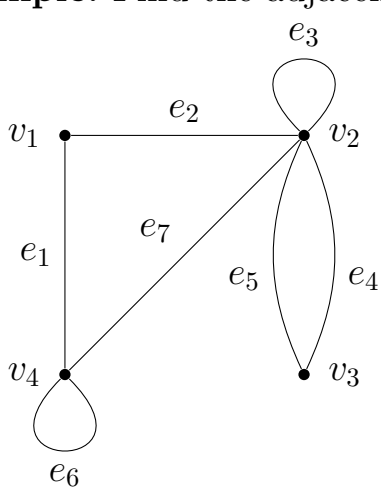
**Example.** Let

$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_3 & v_4 \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 2 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

Draw a directed graph that has $A$ as its adjacency matrix.

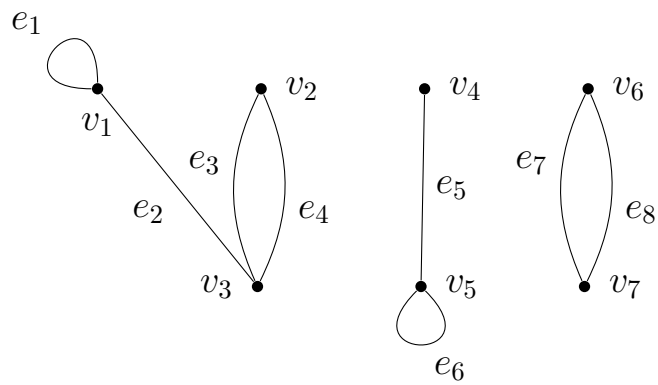



**Example.** Find the adjacency matrix for the following graph:




$$\begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_3 & v_4 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 1 & 2 & 1 \\ 0 & 2 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

**Example.** Find the adjacency matrix of the following disconnected graph:




$$
\begin{array}{c|ccccccc}
 & v_1 & v_2 & v_3 & v_4 & v_5 & v_6 & v_7 \\
\hline
v_1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\
v_2 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\
v_3 & 1 & 2 & 0 & 0 & 0 & 0 & 0 \\
v_4 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
v_5 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\
v_6 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \\
v_7 & 0 & 0 & 0 & 0 & 0 & 2 & 0
\end{array}
$$

**Theorem 10.2.1**
Let $G$ be a graph with connected components $G_1, G_2, \ldots, G_k$. If there are $n_i$ vertices in each connected component $G_i$ and these vertices are numbered consecutively, then the adjacency matrix of $G$ has the form

$$
\begin{bmatrix}
A_1 & O & O & \ldots & O \\
O & A_2 & O & \ldots & O \\
O & O & A_3 & \ldots & O \\
\vdots & \vdots & \vdots & \ddots & \vdots \\
O & O & O & \ldots & A_k
\end{bmatrix}
$$

where each $A_i$ is the $n_i \times n_i$ adjacency matrix of $G_i$, $\forall i = 1, 2, \ldots, k$, and the $O$'s represent matrices whose entries are all 0.

**Definition.**
Suppose that all elements in matrices $A$ and $B$ are real numbers. If the number of elements $n$ in the $i$th row of $A$ equals the number of elements in the $j$th column of $B$, then the **scalar product** or **dot product** of the $i$th row of $A$ and the $j$th column of $B$ is the real number obtained as follows:

$$\begin{bmatrix} a_{i1} & a_{i2} & \dots & a_{in} \end{bmatrix} \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{bmatrix} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

**Example.** Multiple the following matrices:

$$\underbrace{\begin{bmatrix} -2 & -3 & 4 & 5 \end{bmatrix}}_{1\times 4} \underbrace{\begin{bmatrix} -1 \\ 0 \\ 1 \\ -3 \end{bmatrix}}_{4\times 1} = \underbrace{\begin{bmatrix} -9 \end{bmatrix}}_{1\times 1}$$

$(-2)(-1) + (-3)(0) + (4)(1) + (5)(-3)$

$= 2 + 0 + 4 - 15$

**Definition.**
Let $A = (a_{ij})$ be an $m \times k$ matrix, and $B = (b_{ij})$ a $k \times n$ matrix with real entries. The matrix product of $A$ times $B$, denoted $AB$ is the matrix $(c_{ij})$ defined as:

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ik}b_{kj} = \sum_{r=1}^{k} a_{ir}b_{rj}$$

for each $i = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, n$.

**Example.** Multiply the following matrices:

$$\underbrace{\begin{bmatrix} 2 & 0 & -1 \\ -2 & 0 & 2 \end{bmatrix}}_{2 \times 3} \underbrace{\begin{bmatrix} 5 & 5 \\ -2 & -3 \\ 3 & -4 \end{bmatrix}}_{3 \times 2} = \underbrace{\begin{bmatrix} 7 & -4 \\ 14 & -18 \end{bmatrix}}_{2 \times 2}$$

$(2)(5) + 0(-2) + (-1)(3) = 10 + 0 - 3 = 7$

$(2)(5) + 0(-3) + (-1)(-4) = 10 + 0 + 4 = 14$

$(-2)(5) + 0(-2) + (2)(3) = -10 + 0 + 6 = -4$

$(-2)(5) + 0(-3) + (2)(-4) = -10 + 0 - 8 = -18$